Highlights

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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- Gas-lift branch simulated with the two-fluid model, controlled by the platform gas pressure
- Sobol' campaigns from one base state with adaptive horizon: 236 runs over choke and lift gas in two days
- Time-averaged markers at eight points replace trajectories as the training data
- Physics of the averages cuts the residuals two- to four-fold and the gas-rate error with few runs by two to three
- Optimum of revenue minus compression on the surrogate, with capacity constraints on predicted maxima

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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ABSTRACT

A gas-lifted satellite well with a flowline and a riser can produce in severe slugging, in a pressure cycle or in a steady state depending on two operating decisions, the opening of the topside choke and the pressure of the lift gas at the platform, and the time-averaged production of an unstable operating point is not what a steady-state model predicts for it. This paper builds the data and the surrogate models of those two decisions and uses them to optimise the operation of the well. The well, flowline, riser and gas-lift branch are simulated with the four-equation two-fluid model of a companion study, with the flow pattern identified dynamically and the gas-lift line, annulus and check valve simulated with the same model and controlled only by the platform gas pressure. Three sampling campaigns over the two decisions, in rounds of Sobol' points every one of which starts from the final state of the base run, with an adaptive horizon that detects the established regime and captures three cycles, produced 236 settled runs in two days on a laptop over 20 to 260 bar of lift gas, plus 23 budget-limited runs in the corner of high opening and high gas pressure that are re-run with a longer budget; each run is reduced to time-averaged markers (mean, minimum, maximum and the two extreme frequencies) of pressure, void fraction and phase flow rates at eight characteristic points of the network. The map of the decisions shows a stability boundary that runs from 80 bar of lift gas with the choke nearly closed to 125 bar with it open, continuous injection modulated by the cycle below it, and a production gain of 34 % above it at 160 bar and 40 % at 250 bar. Two surrogates of the mean liquid, gas and lift-gas rates and of their spans were trained on the markers: a neural network and a physics-informed network whose physics are the time-averaged mass balances of the well, the gas-lift branch and the valve, exact for the averages and evaluated at collocation points without simulation. With 165 training runs and an early stop set by a scan of the training length, both networks predict the mean liquid rate with a test coefficient of determination of 0.88 to 0.89, the gas and lift-gas rates with 0.996 to 0.998 and the maximum liquid rate with 0.73 to 0.90 depending on the test split; the physics-informed one reduces the residuals of the well and annulus balances two to four times and divides the error of the gas rates by two to three with ten to forty training runs, while the spans of the rates, the signature of slugging, remain the quantity that neither network predicts reliably. The surrogate then enters an operational optimisation of the two decisions: the revenue of the mean liquid rate minus the cost of compressing the lift gas from 5 bar to the platform pressure, with the predicted maxima of the lift-gas and liquid rates bounded by the compressor and separator capacities. With a compressor of 1.75 $\mathrm{kg\,s^{-1}}$ and a separator of 20 $\mathrm{kg\,s^{-1}}$, at the prices of the example, the compression costs one per cent of the revenue and the optimum is the open choke at 229 bar with the compressor as the active constraint; the objective is nearly flat along the constraint, smaller compressors move the optimum down the valve law to the edge of the stable region, and the trade-off against the suction pressure is left for the next step.

1. Introduction

Continuous gas lift is the artificial-lift method of most deepwater satellite wells in Brazil, and its optimisation — how much gas to inject in each well of a field that shares a compressor — has a large literature built on steady-state performance curves (Souza, 2010). A well with a flowline and a riser, however, does not always have a steady state

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to be optimised: with the choke open and little gas it produces in severe slugging (Schmidt, Brill and Beggs, 1980; Taitel, 1986; Jansen, Shoham and Taitel, 1996), and the time-averaged production, the liquid surges that reach the separator and the pressure swings at the wellhead are quantities of the cycle, not of the unstable steady state that a nodal model would return for the same point. A companion study (Souza and Costa, 2026) developed a four-equation two-fluid model with dynamic flow-pattern identification that reproduces the cycle in a well–flowline–riser system of field dimensions, located the onset of slugging as the choke opens and showed, with a gas-lift branch added to the system, that injection at 100 to 120 bar regularises the cycle and from 130 bar suppresses it.

This paper is about the operation of that gas-lifted well. Its question is practical: given the two decisions an operator has, the opening of the topside choke and the pressure of the lift gas at the platform, what does the well deliver on average, with what surges and in which regime — and can a model cheap enough for optimisation answer that over the whole plane of the two decisions? The dynamic simulation is too expensive to be called inside an optimiser (minutes per point), so the paper builds a surrogate, and it does so with a method that follows the companion work on compositional steady-state surrogates (Carvalho, Costa, Nunes and Souza, 2026): a multilayer perceptron trained on simulated data, and the same network with a physics term (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021). The novelty is on both sides of the surrogate. On the data side, the targets are not profiles but time-averaged signatures of a dynamic simulation, and producing them in affordable time required a campaign design of its own: an adaptive horizon that recognises the established regime, warm starts along a shortest path through the sampled points, and a storage that keeps only what the training needs. On the physics side, the governing equations cannot be imposed on a time average of a cycling system; what can, exactly, are the time-averaged mass balances of the well, of the gas-lift branch and of the valve, and the paper shows that this much physics is what the data alone cannot supply.

Section 2 summarises the phenomenological model and the gas-lift branch. Section 3 describes the campaign, the horizon, the markers and the resulting map of the decisions. Section 4 presents the two surrogates and Section 5 compares them. Section 6 formulates the operational optimisation on the surrogate, with capacity constraints on the predicted maxima, and reports its first results.

2. The phenomenological model

The system is the well–flowline–riser of the companion study (Souza and Costa, 2026) with its gas-lift branch (Fig. 1): a vertical well of 1500 m (127 mm) fed by a linear inflow performance relation, a flowline of 2000 m (152.4 mm) whose last 500 m descend at 2° to a low point, a bend of 5 m radius, a riser of 317.5 m and 10 m of horizontal pipe to a choke that discharges into a separator at 10 bar. The gas-lift branch is a 4-in line that runs from the platform along the route of the production line to the wellhead, the casing–tubing annulus (9 5/8 in by 5 1/2 in) to a gas-lift valve 30 m above the bottom of the well, and a check valve fed by a chamber at the base of the annulus. The base well has a reservoir pressure of 200 bar, a productivity index of 5 $\mathrm{m^3\,d^{-1}\,bar^{-1}}$ and a gas–oil ratio of 50 $\mathrm{m^3\,m^{-3}}$; the fluid is the light oil of Carvalho (2022) frozen as a polytropic pair.

Every pipe, the gas pipes included, is described by the one-dimensional four-equation two-fluid model of the companion study: mass and momentum of each phase, closures by flow pattern (stratified, intermittent, bubbly, annular) selected dynamically on the local state through smooth logistic switches, a low-void rule, an objective virtual-mass term and an interfacial pressure in the slug unit that keep the model hyperbolic along the simulated trajectories. The staggered finite-volume discretisation with phase-wise donor cells and the conservative variable-step implicit integration of that study are kept unchanged, with a compiled right-hand side and exact Jacobians; the base mesh has 15 cells in the well, 35 in the flowline–riser pipe, 19 in the gas-lift line and 10 in the annulus, 323 states in all. The gas-lift branch is controlled by one number, the gas pressure P_{gas} at the platform end of the line: the injected rate is not prescribed but follows from P_{gas}, the static gas columns of the line and the annulus, the valve law $q = 0.0075\,C_V\sqrt{\rho_G\,\Delta P}$ ($C_V = 1.5$, ΔP in bar, regularised at the origin) and the pressure of the well at the injection point, so that it responds to whatever the well does; at 20 bar the valve never opens and the system is the base case of the companion study with the branch inert. The reservoir gas is the gas of the well; the lift gas adds to it at the injection point.

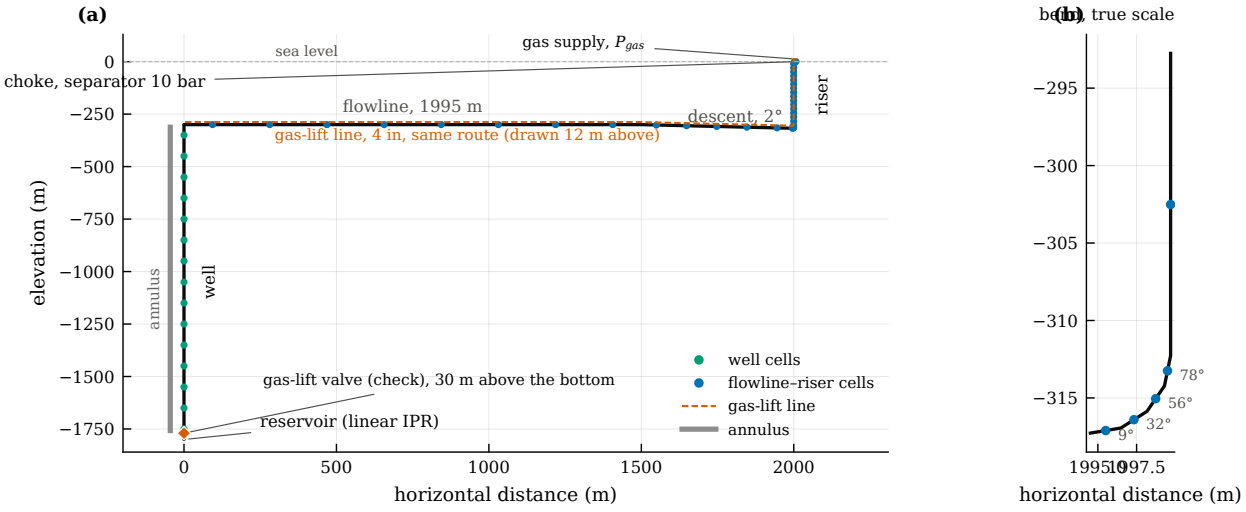


Figure 1: The well–flowline–riser system with the gas-lift branch (schematic, from the companion study): the gas-lift line follows the route of the production line, the annulus runs beside the well and the check valve injects 30 m above the bottom. The two operating decisions are the choke opening a and the platform gas pressure P_{gas}.

3. Data: campaign, horizon and markers

3.1. Sampling the two decisions

The inputs are the choke opening $a \in [0.1, 1]$ and the platform gas pressure P_{gas}, with the well fixed at its base state, in three campaigns: a first over 20 to 160 bar, a second over 160 to 260 bar once the first had placed the optimum of Section 6 on its upper edge, and a third over 160 to 260 bar restricted to openings of 0.5 to 1, the region of that optimum. Points are drawn from a scrambled Sobol' sequence (Sobol', 1967; Owen, 1998) in rounds of 25, so that the rounds of a campaign together remain a single space-filling design, and each point is identified by a hash of its input tuple; only what the training needs is stored, the markers below and the final state of the run. The initial condition of every run is the same: the final state of the base run of the companion study, 8000 s of severe slugging with the choke open and the gas line at 20 bar. The campaigns began with a different rule, one chain per round in which the first point started from the base state and every later point from the final state of the previous one, along the shortest open path through the 25 points on the inputs rescaled to $[-1, 1]$; the rule spares the transient of a cold start, but above an opening of 0.5 with more than 100 bar of lift gas every run started from a neighbour exhausted its budget while every run started from the base state settled in about a minute, and the campaigns were completed from the base state. A steady-state initial condition, the natural alternative, is not attainable for this model: its steady equations are discontinuous in the state, because the closures switch flow pattern at the base of the riser, and a Newton iteration converges only from a state that already has the right pattern in each cell, which a settled dynamic run provides and no algebraic estimate does. The transient from the base state costs 160 to 2000 steps where the well is steady and up to 16000 where it cycles, and it does not depend on the order of the points, so the runs of a round are distributed over processes without any sequencing.

3.2. Adaptive horizon with capture of cycles

Each run integrates in windows of 3000 s and monitors two signals, the pressure at the wellhead and the liquid rate through the choke. After a minimum time (4000 s from a blocked start, 1500 s from a warm start) the regime is taken as established when both signals are permanent (amplitude below 0.1 bar and 0.05 kg s^{-1}) or when two consecutive windows have the same envelope, that is, the same amplitude, mean, minimum and maximum within 10 % of the amplitude; the envelope criterion, rather than equality of periods, is what accepts the cycles whose blowouts alternate in size. When a monitored signal cycles with a period that fits ten times in the window, the comparison window shrinks to ten periods (at least 600 s) and is evaluated on the series already integrated, so that a fast cycle is accepted about 1200 s after it appears instead of two full windows later. The run then continues for three more cycles of the longest monitored period (600 s for a steady state), which are the capture window of the markers, and stops; a run that never

settles stops at 24 000 s and is kept with the markers of its last 4000 s and a flag. A wall-clock budget of 40 min per run, raised to 60 min for the re-runs of the points that had exhausted it, bounds the points where the integrator crawls: with the choke more than half open and more than 100 bar of lift gas the liquid outflow at the top of the riser oscillates with a period under a minute and an amplitude of up to 15 kg s^{-1}, and the steps fall to 0.3 to 0.6 s with a quarter of them rejected; the short horizontal section between the riser and the choke, the open point of the closures that the companion study reported, is not the cause, since the same runs without it cost the same. Those runs are flagged and excluded from the training.

3.3. Markers at characteristic points

A run is reduced to markers of its capture window at eight points of the network: the bottom of the well, the injection point, the wellhead, the low point of the flowline, the top of the riser, the two ends of the gas-lift line and the base of the annulus. At each point four signals are recorded — pressure, void fraction, liquid and gas mass flow rates — and each signal is summarised by its mean, minimum, maximum and two frequencies, the inverse of the longest and of the shortest interval between successive maxima of the smoothed signal: equal for a periodic cycle, different when blowouts alternate, zero for a permanent signal. The gas injected through the valve is recorded in the same way, and a class is assigned from the wellhead pressure and the choke liquid rate as in the companion study (steady, pressure cycle or severe slugging when the peak liquid rate exceeds twice its mean), with a fourth class, the fast cycle, when the pressure cycles with a period under 300 s: it is the hydrodynamic intermittency of the riser top of Section 3.2, which the peak criterion alone would read as severe slugging although the wellhead pressure swings by a fraction of a bar; the settled severe-slugging and pressure cycles of the campaigns all have periods above 540 s. Trajectories are not kept: they would cost 12 MB per run against 10 kB of markers, and the surrogate consumes only the markers.

3.4. The map of the decisions

Table 1 summarises the three campaigns and Fig. 2 shows them on the plane of the decisions. Of the 236 runs that settled, 101 produce in severe slugging, 13 in a pressure cycle, 3 in the fast cycle and 119 in a steady state; 120 settled on a repeated cycle, 114 on a steady state and 2 reached the time limit; 23 further runs exhausted the budget of 40 min and were re-run with 60 (below). The stability boundary is a diagonal that runs from about 90 bar of lift gas with the choke at 0.3 to 100 to 125 bar with it open: the more open the choke, the more gas is needed to suppress the cycle, and at the smallest openings some throttled points are steady without lift gas. The check valve opens between 86 and 94 bar almost independently of the choke, so that below 90 bar the data are the base well with the choke as the only decision; between the opening of the valve and the boundary the injection is continuous and modulated by the cycle, which shortens and regularises; above the boundary the well is steady and its production rises with the gas, to 5.3 kg s^{-1} at 160 bar with the choke open, 34 % above the base case, and on to 5.6 kg s^{-1} at 250 bar with the choke half open, a tenth of the gain per bar of the first hundred bar above the boundary. The peak liquid rate through the choke reaches 45 kg s^{-1} in the slugging region with the choke open and stays below 6 kg s^{-1} in the steady region. The 23 runs that exhausted the budget sit in a strip of openings from 0.5 to 0.99 between 106 and 244 bar, where the riser receives the most gas and the liquid outflow cycles fast (Section 3.2). Twenty-two of them have been re-run from the base state with the longer budget: seven settled in a steady state within three minutes (their first failure was the warm start of the early campaigns), four settled on cycles after 37 to 57 minutes (three of them fast cycles accepted by the shortened window, at up to 0.1 s per step), and eleven exhausted the hour as well, at 0.04 to 0.1 s per step: three on the stability boundary with the choke nearly open, in severe slugging, and eight on the fast cycle of the corner, whose class the markers of their partial trajectory already give. None enters the training until it settles, and they mark where the simulator, not the well, is least efficient.

4. Surrogates

4.1. Outputs and network

The surrogate maps the two decisions to eleven quantities of the capture window: the mean liquid and gas rates at the top of the riser, the mean gas-lift rate through the valve, the spans (maximum minus minimum) of the same three rates, which are the signature of slugging, the maxima of the liquid rate at the top and of the gas-lift rate, which the capacity constraints of the optimisation use, and three auxiliary mean pressures — at the bottom of the well, at the injection point and at the base of the annulus — that the physics of the next section needs. Spans and maxima are learnt as $\log(1 + x)$, so that they are never negative and their orders of magnitude are resolved. The network is a multilayer

Table 1
The sampling campaign over the two decisions.

points (runs)	247
steady / fast cycle / pressure cycle / severe slugging	130 / 3 / 13 / 101
stop: steady / cycle captured / time limit / budget	114 / 120 / 2 / 11
simulated time per run (s): median, min–max	9600, 3505–24000
accepted steps per run: median, min–max	6073, 159–81742
integration time per run (min): median, max	3.9, 60.0

Base well of Part I ($P_{res} = 200\,\mathrm{bar}$, $J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, GOR 50) with the gas-lift branch; inputs sampled by scrambled Sobol' sequences in rounds of 25 over $a \in [0.1, 1]$ and $P_{gas} \in [20, 160]$ bar (first campaign), [160, 260] bar (second) and [160, 260] bar with $a \in [0.5, 1]$ (third, densifying the region of the optimum); each round starts from the final state of the base run at the point closest to it and continues along the shortest path with warm starts. Budget: 40 min of integration per run.

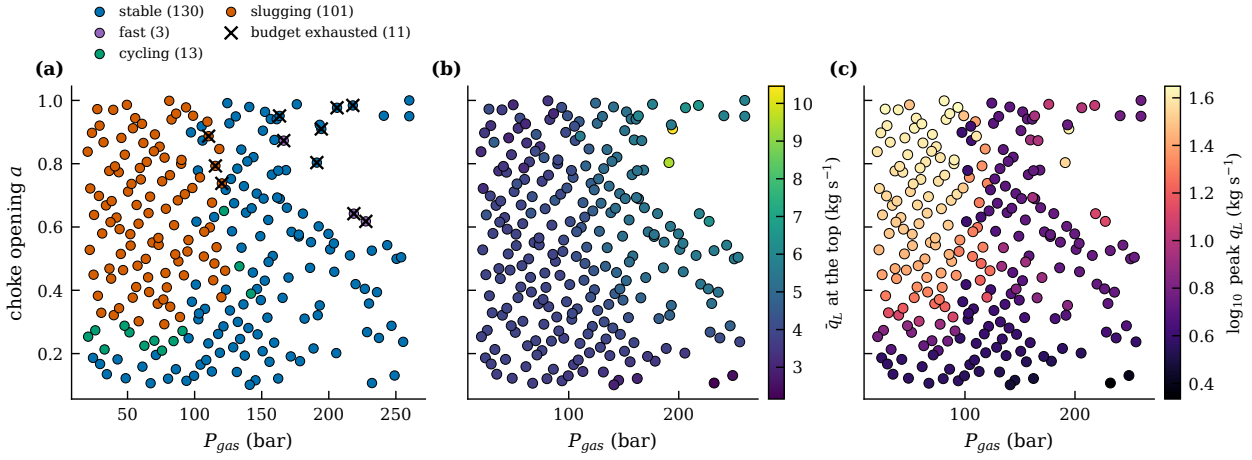


Figure 2: The 247 runs on the plane of the choke opening and the platform gas pressure (the three campaigns and the verification runs of Section 6 that settled): (a) class of the established regime, with the eleven runs that exhausted the 60-min integration budget before settling (crosses; their markers come from a partial trajectory and they are excluded from the training); (b) mean liquid rate through the choke; (c) peak liquid rate through the choke on a logarithmic scale.

perceptron (Hornik, Stinchcombe and White, 1989) with two hidden layers of 32 hyperbolic-tangent units and a linear output, inputs and outputs standardised with the training statistics, trained with Adam (Kingma and Ba, 2015) in full batch for 2000 epochs on the mean squared error of the standardised outputs, with the epochs unrolled in a compiled scan so that a network trains in a few seconds; 70 % of the 236 settled runs train it and 30 % test it. The number of epochs is the early stop of a scan from 1000 to 16000: the test error of the mean liquid rate is flat between 1000 and 2000 epochs and rises steadily beyond, for both networks, as the fit of the training runs approaches $R^2 = 1$ — with 150 runs and 1200 weights the training time is the regularisation, and the physics-informed network degrades more slowly than the data-only one. A grid of one or two layers, 8 to 64 units and three learning rates (24 networks of each kind, Section 5) checked that the architecture is not critical.

4.2. Physics of the averages

A physics-informed network adds to the data loss the residual of the governing equations at collocation points that need not be simulated (Raissi et al., 2019). Here the targets are time averages of a dynamic system, and the time average of a cycle does not satisfy the two-fluid equations, which are nonlinear; what it satisfies exactly are the balances that are linear in the averaged quantities. Over a periodic cycle or a steady state, the mass of each phase in the system does not change on average, so that

$$R_1 = \bar{q}_L - J_L\,(P_{res} - \bar{P}_{bh}), \tag{1}$$

$$R_2 = \bar{q}_G - J_G\,(P_{res} - \bar{P}_{bh}) - \bar{q}_{gl}, \tag{2}$$

$$R_3 = \bar{P}_{an} - \kappa\, P_{gas}, \tag{3}$$

$$R_4 = \bar{q}_{gl} - 0.0075\, C_V \sqrt{\rho_G(\bar{P})\, \langle \bar{P}_{an} - \bar{P}_{inj} \rangle_+}, \tag{4}$$

where R_1 and R_2 are the liquid and gas balances between the linear inflow relation at the mean bottom-hole pressure and the mean rates at the top, R_3 is the static gas column from the platform to the base of the annulus ($\kappa = 1.138$ for this fluid and depth), and R_4 is the valve law written with the mean pressures, the only approximate relation of the four (the mean of a square root is not the square root of the mean, but the modulation of the valve pressure drop is within 25 % and the error is a few per cent), with a smooth gate that closes the valve when the mean annulus pressure falls below the mean injection pressure. The loss is the data term plus λ times the mean square of the four residuals, each scaled by the standard deviation of the quantity it compares, evaluated on 200 Latin-hypercube points (McKay, Beckman and Conover, 1979) of the plane of the decisions; $\lambda = 1$, and values from 0.1 to 100 give the same fit with residuals decreasing with λ. The residuals are closed-form functions of the network outputs, so that the gradient of the physics term is exact and cheap, unlike the detached-Jacobian construction that the compositional surrogates needed (Carvalho et al., 2026). The information in the physics is therefore not the dynamics but the accounting: the network is told that whatever enters the well on average leaves it on average, and that the lift gas is set by two mean pressures it also predicts.

5. Results

Table 2 and Fig. 3 compare the two networks on the 71 test runs. Both networks predict the mean gas rate at the top and the mean gas-lift rate with R^2 of 0.996 to 0.998 and the three mean pressures with 0.97 to 1.00; the mean liquid rate, 0.88 for the data-only network and 0.89 for the physics-informed one (root-mean-square errors of 0.25 and 0.23 kg s^{-1}), is the hardest of the means: it varies little over the plane (2.2 to 5.6 kg s^{-1}), its time average over three cycles of a surging signal is the noisiest of the targets, and the third campaign, dense in the steady corner, adds points where it changes by a few per cent across the plane. With the first campaign alone (105 training runs) the data-only network reached only 0.57 on it and the physics raised it to 0.88; with the three campaigns and the early stop the two networks are equal on every mean, and the physics show in the balances: the scaled residuals of R_1 (liquid inflow) and R_3 (annulus column) fall by factors of four and two, while R_2 and R_4, which both networks satisfy to within 0.04 standard units, do not separate them. The maxima behave like the means they ride on: the maximum lift-gas rate reaches 0.996 with both networks, since the injection is at most modulated by a quarter, and the maximum liquid rate, which carries the surge, 0.73 to 0.74 on this split and between 0.73 and 0.92 across the successive splits of this study. The spans are the weak point of both: the liquid span scores between negative and 0.9 depending on which runs fall in the test set (the data-only network fails outright here, the physics-informed one scores 0.54), and the gas span and the span of the injected gas stay below 0.6, since the amplitude of a surge in three cycles is a noisy marker by nature. Fig. 4 shows the physics-informed surrogate on the plane of the decisions: the mean liquid rate rises with the lift gas above the boundary and falls as the choke closes, the lift-gas rate is zero below 90 bar and rises with P_{gas} as the valve law dictates, and the maximum liquid rate marks the slugging region, with surges of 20 to 40 kg s^{-1} below the boundary against 5 above it, and a narrow band of artefacts along the boundary, where the data change class within a few bar.

With few data the physics counts more. Table 3 and Fig. 5 retrain both networks on 10, 20 and 40 runs drawn from the training set, three draws each, and test them on the same 65 runs. The physics-informed network has the lower error for the gas and lift-gas rates at every size — a third to a half of the error of the data-only network with 10, 20 and 40 runs — and for the liquid rate at every size, 0.37, 0.31 and 0.28 kg s^{-1} against 0.53, 0.40 and 0.36; with 40 runs its gas error is twice what it reaches with 165 (0.032 against 0.037 kg s^{-1}) and its liquid error 0.28 against 0.23. The hyperparameter grid confirms that the architecture is not what limits the fit: all 24 architectures of each kind reach a mean R^2 of 0.8 over the six requested outputs, and the differences between architectures are smaller than the difference the physics makes with few data.

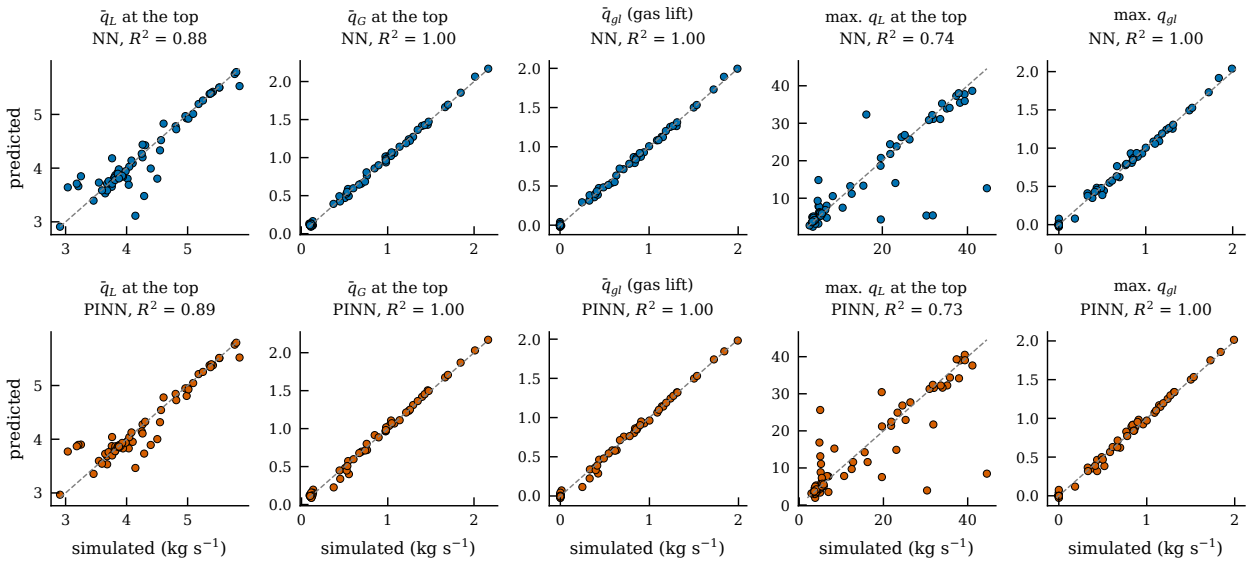


Figure 3: Test parity of the mean liquid, gas and lift-gas rates and of the maxima of the liquid and lift-gas rates, for the data-only network (top row) and the physics-informed network (bottom row).

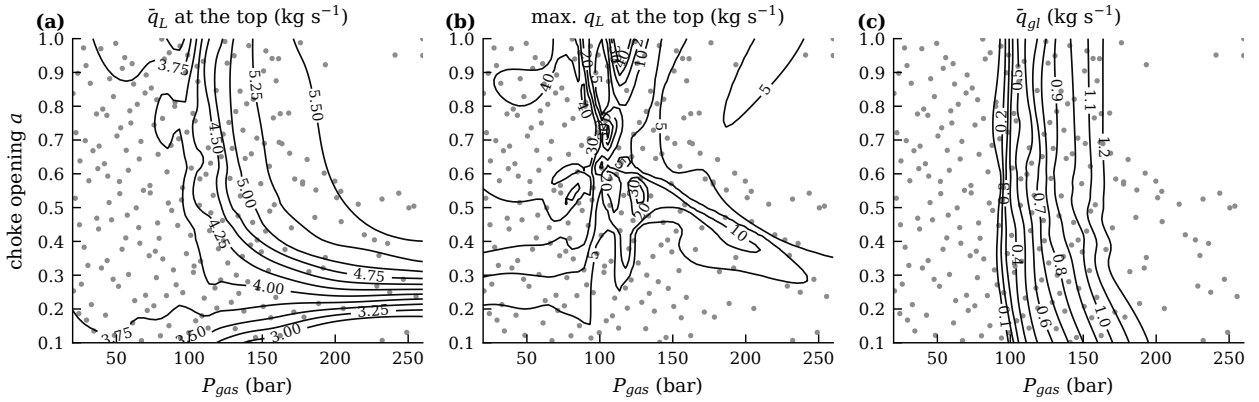


Figure 4: The physics-informed surrogate on the plane of the decisions as labelled level curves, with the positions of the simulated runs (grey dots): (a) mean liquid rate at the top; (b) maximum liquid rate at the top, the surge that reaches the separator; (c) mean gas-lift rate.

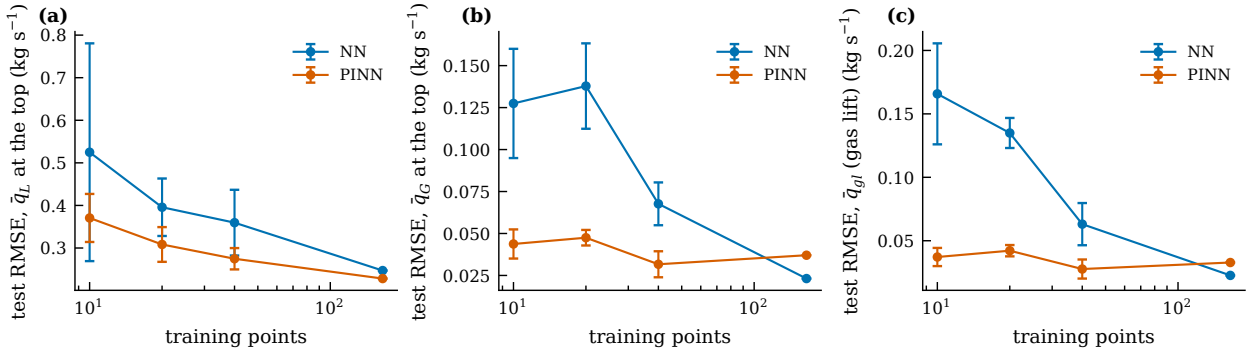


Figure 5: Learning curves of the two surrogates: test root-mean-square error of (a) the mean liquid rate, (b) the mean gas rate and (c) the mean gas-lift rate against the number of training runs (mean and standard deviation over three draws; the last point is the full training set).

Table 2
Test metrics of the two surrogates.

output	NN R^2	NN RMSE	PINN R^2	PINN RMSE
$\bar{q}_L$ at the top	0.876	0.247	0.894	0.228
$\bar{q}_G$ at the top	0.998	0.023	0.996	0.037
$\bar{q}_{gl}$ (gas lift)	0.998	0.023	0.996	0.033
span of q_L	-1.127	21.771	0.540	10.127
span of q_G	-0.397	0.868	0.001	0.734
span of q_{gl}	0.378	0.056	0.542	0.048
max. q_L at the top	0.740	6.658	0.726	6.832
max. q_{gl}	0.995	0.038	0.996	0.037
$\bar{P}_{bh}$	0.992	1.404	0.969	2.759
$\bar{P}_{inj}$	0.992	1.396	0.986	1.844
$\bar{P}_{an}$	0.998	2.908	0.999	2.081
residuals R_1–R_4 (RMS, std units)	–	0.140, 0.004, 0.045, 0.037	–	0.039, 0.017, 0.025, 0.031

Test set of 71 runs (165 for training); RMSE in kg/s for the flow rates and bar for the pressures. Residuals: root-mean-square of the four scaled physics residuals on 400 held-out collocation points. Both networks have two hidden layers of 32 hyperbolic-tangent units; the PINN adds the physics term with $\lambda = 1$ on 200 Latin-hypercube collocation points.

Table 3
Sparse-data study: test error of the mean rates against the number of training runs.

training points	$\bar{q}_L$ NN	$\bar{q}_L$ PINN	$\bar{q}_G$ NN	$\bar{q}_G$ PINN	$\bar{q}_{gl}$ NN	$\bar{q}_{gl}$ PINN
10	0.525 ± 0.256	0.371 ± 0.056	0.127 ± 0.033	0.044 ± 0.009	0.166 ± 0.040	0.037 ± 0.007
20	0.396 ± 0.067	0.308 ± 0.041	0.138 ± 0.025	0.047 ± 0.005	0.135 ± 0.012	0.042 ± 0.004
40	0.360 ± 0.077	0.275 ± 0.025	0.068 ± 0.013	0.032 ± 0.008	0.063 ± 0.017	0.028 ± 0.008
165	0.247	0.228	0.023	0.037	0.023	0.033

Test RMSE (kg/s), mean ± standard deviation over three random subsets of the training set; last row: the full training set.

6. Operational optimisation

6.1. Formulation

The operator chooses the choke opening a and the platform gas pressure P_{gas} to maximise the daily margin of the well,

$$F(a, P_{gas}) = c_{oil}\, Q_o(\bar{q}_L) - c_e\, 24\, W(\bar{q}_{gl}, P_{gas}), \tag{5}$$

the revenue of the oil minus the cost of compressing the lift gas, subject to the capacities of the compressor and of the separator,

$$\max q_{gl} \le Q_{gl}^{cap}, \qquad \max q_L \le Q_L^{cap}, \tag{6}$$

written on the maxima of the rates over the cycle, which the surrogate predicts (Section 4.1), and not on their means: a compressor and a separator are sized for peaks. The oil rate Q_o follows from the mean liquid rate with the liquid mass per stock-tank cubic metre of the base fluid (700 kg m^{-3}, oil and solution gas); the compression power W is that of a polytropic compression of the mean lift-gas rate from the suction pressure (5 bar) to P_{gas} in stages of ratio at most four with intercooling to the suction temperature ($k = 1.3$, 313 K, efficiency 0.75, molar mass 19.5 g mol^{-1} from the gas density of the fluid). All the quantities in Eqs. (5) and (6) come from the physics-informed surrogate, so that F and the constraints are smooth closed-form functions of the two decisions, evaluated in microseconds. The problem is solved by enumeration on a 201 × 201 grid of the plane restricted to the feasible nodes, followed by a sequential quadratic programming refinement with the two constraints and the bounds, and the optimum is confirmed by one run of the simulator at that point.

6.2. First results

[Seed of the optimisation study; the Pareto front against the suction pressure is pending.] The example takes a compressor of 1.75 kg s^{-1}, a separator that accepts peaks of 20 kg s^{-1}, an oil price of 70 \$ per barrel and an energy

price of 0.10 \$ per kW h. Fig. 6 shows the objective on the plane of the decisions with the runs of the campaigns by class, the level curves of the two predicted peaks and the optimum: the open choke at $(a, P_{gas}) = (1.0, 229\,\text{bar})$, with a margin of 309×10^3 \$/d, a revenue of 312×10^3 \$/d against a compression cost of 3.3×10^3 \$/d (1375 kW for a mean lift-gas rate of $1.74\,\text{kg s}^{-1}$). The compressor is the active constraint and the separator is not, since the predicted peak liquid rate there is $4.8\,\text{kg s}^{-1}$. The compression is one per cent of the revenue, so the objective is the production, which rises with the lift gas along the valve law until the compressor stops it; along the constraint the objective is nearly indifferent to the opening, losing 0.7 % at 0.9 and 3 % at 0.5, so the choke stays open, and the open choke is the physical bound, not the edge of the sampled box. The simulator run at (1.0, 224 bar), 5 bar below the optimum on the constraint and started from the base state, reproduces the surrogate in the means: over its last 4000 s the liquid rate is 5.79 against the predicted $5.74\,\text{kg s}^{-1}$, the gas rate 1.88 against 1.89, the lift-gas rate 1.71 against 1.70, the bottom-hole pressure 58.4 against 58.8 bar and the annulus pressure 254 against 256. It does not reproduce the maximum: the run sits on the fast cycle of the riser top described in Section 3.2, with the liquid rate through the choke swinging between 2 and $16\,\text{kg s}^{-1}$ every minute while the wellhead pressure stays within 0.3 bar, and the surrogate, trained on the runs that settled, predicts a peak of 4.8. The optimum therefore sits in the strip of budget-limited runs of Section 3.4 for a physical reason, the fast cycle, not for a numerical one, and the objective, which depends on the means, is confirmed to 1 %; a separator constraint written on the peak would need the fast cycle in the training data, which asks for a budget of about 60 min per run in that strip (the re-runs of Section 3.4). A second run on the plateau, at (0.94, 205 bar), agrees with the surrogate in the means and pressures to better than 1.5 %, and there the fast cycle has an amplitude of only $0.6\,\text{kg s}^{-1}$, so its peak (5.9 against a predicted 4.7) is closer; a third run at (0.95, 260 bar), the far corner of the box, agrees to 0.5 % in every mean and pressure and to 5 % in the peak (5.8 against 5.5), and is steady: the amplitude of the fast cycle varies across the strip. At 260 bar with the choke open the same well is steady, and stays steady when its run is restarted from the settled state. The optimum is insensitive to the oil price, since the compression stays one per cent of the revenue (at 20 \$ per barrel it moves only to 228 bar), and leaves the plateau only through the energy price: at 2 \$ per kW h the compression is a seventh of the revenue and the optimum retreats to 173 bar, and at 5 \$ per kW h it jumps below the opening of the valve, (0.77, 85 bar), the best point of the well without lift gas, since the margin of the lift gas no longer pays its compression.

The constraints change the picture as soon as they bind, and the compressor binds first. The valve law makes the lift-gas rate a function of P_{gas} alone, so a compressor capacity fixes the platform pressure and the optimum walks down the valve law as the capacity shrinks, with a separator that accepts peaks of $20\,\text{kg s}^{-1}$ never active: with $1.5\,\text{kg s}^{-1}$ the optimum is (1.0, 196 bar) with a margin of 304×10^3 \$/d, two per cent below, since the objective is flat along the valve law at that level; with $1.0\,\text{kg s}^{-1}$ it is 142 bar and 282×10^3 \$/d, 9 % below; with $0.8\,\text{kg s}^{-1}$ it is 122 bar and 259×10^3 \$/d, 16 % below and just above the stability boundary, with a predicted peak liquid rate of $5.5\,\text{kg s}^{-1}$. Along the constraint the objective is nearly indifferent to the opening above 0.7 (it loses one per cent at 0.9 and three to six per cent at 0.5, and successive fits of the surrogate place the optimum anywhere between 0.7 and 1), so the choke stays open. A smaller compressor pushes the optimum across the boundary: with $0.5\,\text{kg s}^{-1}$ it goes to about 103 bar, in the slugging region, with surges of $12\,\text{kg s}^{-1}$ that a separator of 20 still accepts; with a separator of $6\,\text{kg s}^{-1}$ the separator constraint takes over and holds the optimum at 106 bar, at the edge of the cycle, which is what a constraint written on the peak, not on the mean, is for.

7. Conclusions

A dynamic two-fluid simulation of a gas-lifted well can feed a surrogate of its operating decisions once the campaign is designed for it: one fixed initial condition for every point, the final state of the base run, which the integrator of this model tolerates where a warm start from a neighbour or a Newton steady state does not, an adaptive horizon that recognises the established regime and captures three cycles, and markers at characteristic points instead of trajectories. Over the plane of the choke opening and the platform gas pressure the well has a stability boundary from 90 to 125 bar of lift gas, continuous injection modulated by the cycle below it and 34 % more production above it at 160 bar, 40 % at 250 bar. Of the two surrogates trained on the 230 runs, the physics-informed one, whose physics are the time-averaged mass balances of the well and of the gas-lift branch, predicts the mean liquid rate with R^2 0.89 against 0.88, the gas and lift-gas rates with 0.996 to 0.998 and their maxima with 0.73 to 0.90 (across the test splits of this study) and 0.996, satisfies the well and annulus balances two to four times better than the data-only network, needs a third to a half of the data for the same error on the gas rates, and leaves the spans of the rates as the quantity to be handled by the class and the stability map rather than by regression. On this surrogate the optimisation of the revenue minus the compression

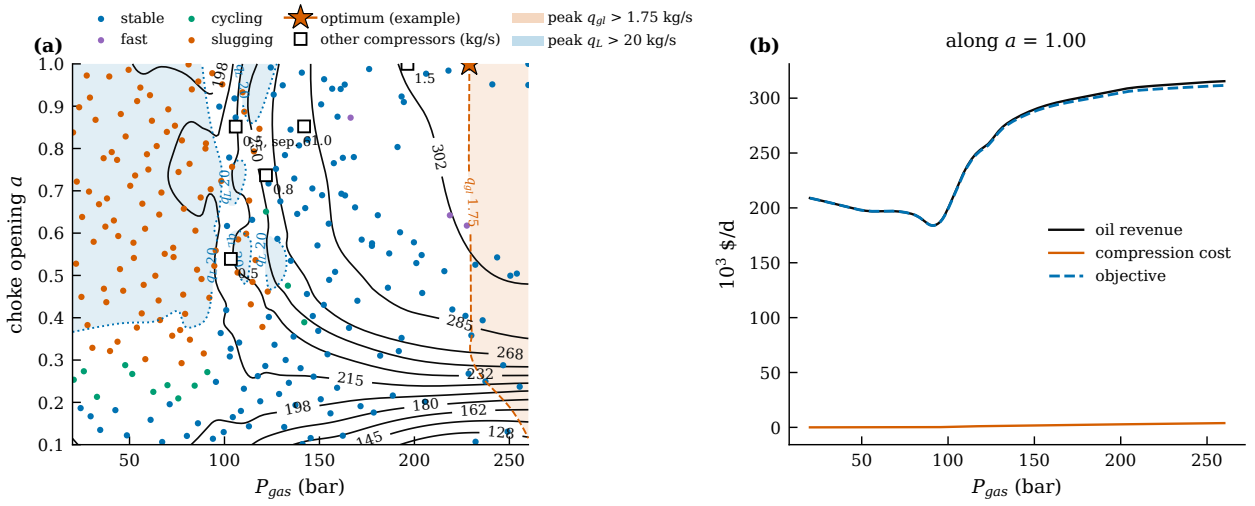


Figure 6: Operational optimisation on the surrogate at 70 $ per barrel and 0.10 $ per kW h: (a) objective of Eq. (5) on the plane of the decisions (level curves, 10^3 $/d), the runs of the campaign by class, the optimum of the example (compressor of 1.75 kg s^{-1}, separator of 20; star), the two constraints as level curves and shaded infeasible regions, the predicted peak lift-gas rate above 1.75 kg s^{-1} (dashed, red) and the predicted peak liquid rate above 20 kg s^{-1} (dotted, blue), and the optima for the other compressor capacities of the text (squares, labelled in kg s^{-1}; the last one also with a separator of 6); (b) revenue, compression cost and objective along the platform gas pressure with the choke at the optimal opening, without the constraints.

cost, with the predicted maxima bounded by the compressor and separator capacities, is a two-variable problem solved by enumeration and refined in a fraction of a second: with a compressor of 1.75 kg s^{-1} and a separator of 20 kg s^{-1}, at the prices of the example, the compression is one per cent of the revenue and the optimum is the open choke at 229 bar with the compressor as the active constraint; the objective is nearly flat along the valve law, a smaller compressor moves the optimum down the platform pressure to the stability boundary, and a separator capacity written on the peak liquid rate is what keeps it out of the cycle. The trade-off against the suction pressure, as a Pareto front, is the next step.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for language editing and code assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator used in this study is not publicly available; the campaign design (sampling, horizon, markers) and the surrogates are described in full so that they can be reproduced with an independent implementation, and the table of markers and the trained networks are available from the corresponding author upon reasonable request.

References

Carvalho, G.A.d., 2022. Modelagem e simulação composicional do escoamento multifásico em tubulações com geração de modelos substitutos. Dissertação de mestrado. Universidade do Estado do Rio de Janeiro, Instituto de Química. Rio de Janeiro.

Carvalho, G.A.d., Costa, A.L.H., Nunes, G.C., Souza, J.N.M.d., 2026. Steady-state compositional gas–liquid flow in production pipelines. Part II: neural and physics-informed surrogates with the axial position as a feature. Companion paper, submitted to Geoenergy Science and Engineering.

Hornik, K., Stinchcombe, M., White, H., 1989. Multilayer feedforward networks are universal approximators. Neural Netw. 2, 359–366. doi:10.1016/0893-6080(89)90020-8.

Jansen, F.E., Shoham, O., Taitel, Y., 1996. The elimination of severe slugging—experiments and modeling. Int. J. Multiph. Flow 22, 1055–1072. doi:10.1016/0301-9322(96)00027-4.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nat. Rev. Phys. 3, 422–440. doi:10.1038/s42254-021-00314-5.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization, in: 3rd International Conference on Learning Representations (ICLR), San Diego, CA. ArXiv:1412.6980.

McKay, M.D., Beckman, R.J., Conover, W.J., 1979. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. Technometrics 21, 239–245. doi:10.1080/00401706.1979.10489755.

Owen, A.B., 1998. Scrambling Sobol' and Niederreiter–Xing points. J. Complexity 14, 466–489. doi:10.1006/jcom.1998.0487.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707. doi:10.1016/j.jcp.2018.10.045.

Schmidt, Z., Brill, J.P., Beggs, H.D., 1980. Experimental study of severe slugging in a two-phase-flow pipeline–riser pipe system. Soc. Pet. Eng. J. 20, 407–414. doi:10.2118/8306-PA.

Sobol', I.M., 1967. On the distribution of points in a cube and the approximate evaluation of integrals. USSR Comput. Math. Math. Phys. 7, 86–112. doi:10.1016/0041-5553(67)90144-9.

Souza, J.N.M., 2010. Modelagem e simulação de escoamento multifásico em dutos de produção de óleo e gás natural. D.Sc. thesis. Escola de Química, Universidade Federal do Rio de Janeiro. Rio de Janeiro, Brazil. In Portuguese.

Souza, J.N.M.d., Costa, A.L.H., 2026. Severe slugging with a four-equation two-fluid model and dynamic flow-pattern identification: well-posedness, discretisation, time integration and choke stability maps. Companion paper (Part I of this study), submitted.

Taitel, Y., 1986. Stability of severe slugging. Int. J. Multiph. Flow 12, 203–217. doi:10.1016/0301-9322(86)90026-1.